

MISSION 100

JEE 2027

Lecture – 02

Physics

Mathematical Tools and Vector

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Recap of Previous Lecture

1

Vector Basics

2

Resolution of Vectors

3

Vector Addition

4

Vector Multiplication



Topics to be Covered

Cross Product

1

Differentiation

2

Maxima Minima

3

Integration

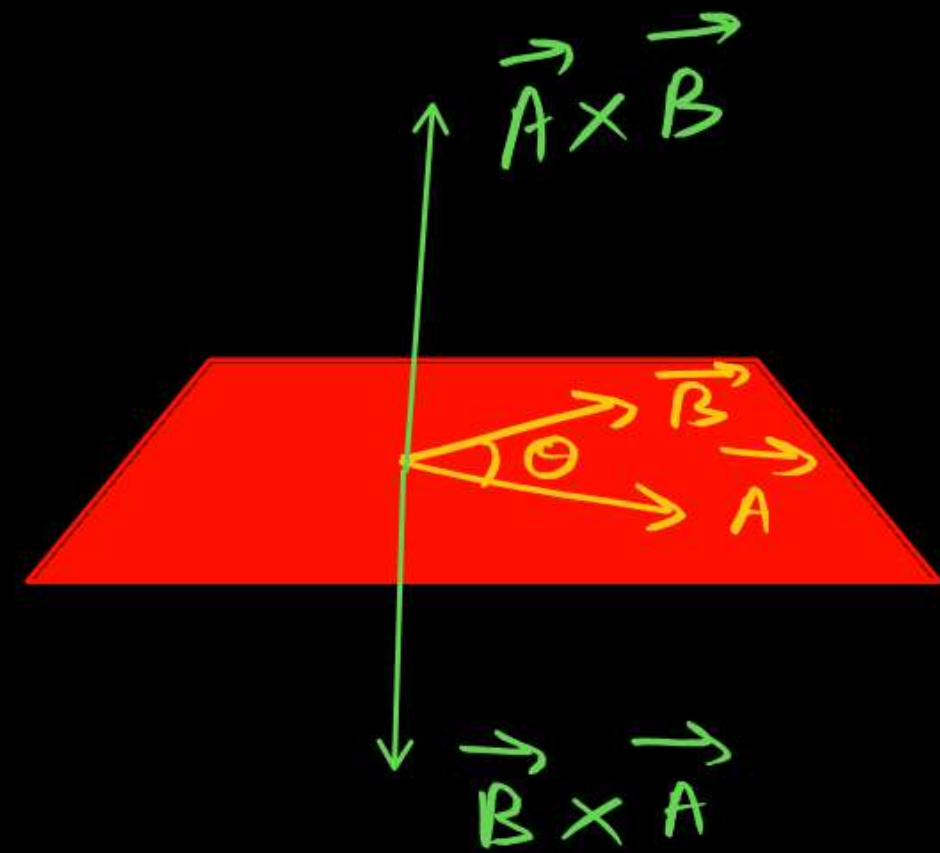
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Graphs

Cross Product (Vector Product):

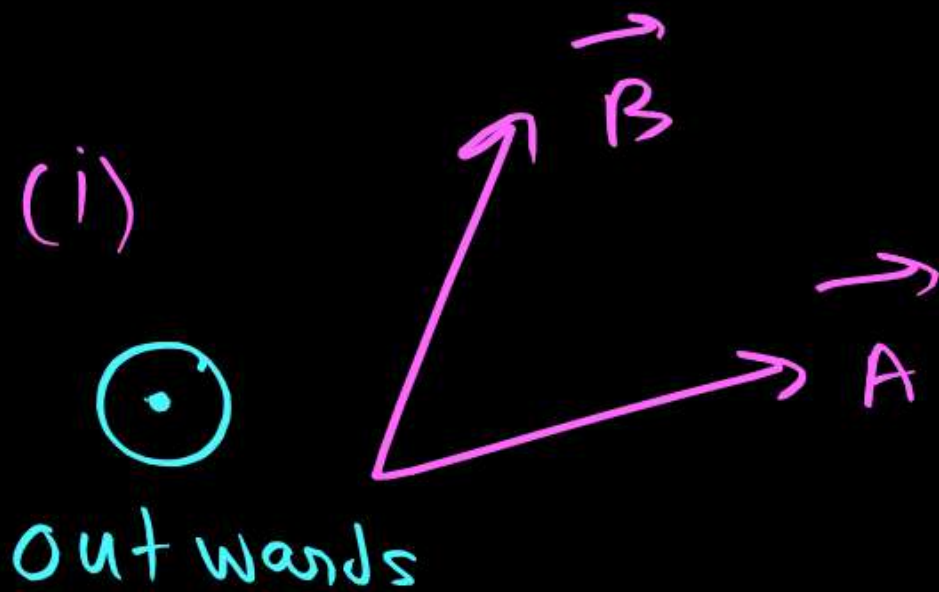
$$\vec{A} \times \vec{B} = \underline{|\vec{A}| |\vec{B}| \sin \theta} (\hat{n})$$

$\hat{n} \rightarrow$ unit vector $\perp$ to both $\vec{A}$ & $\vec{B}$.

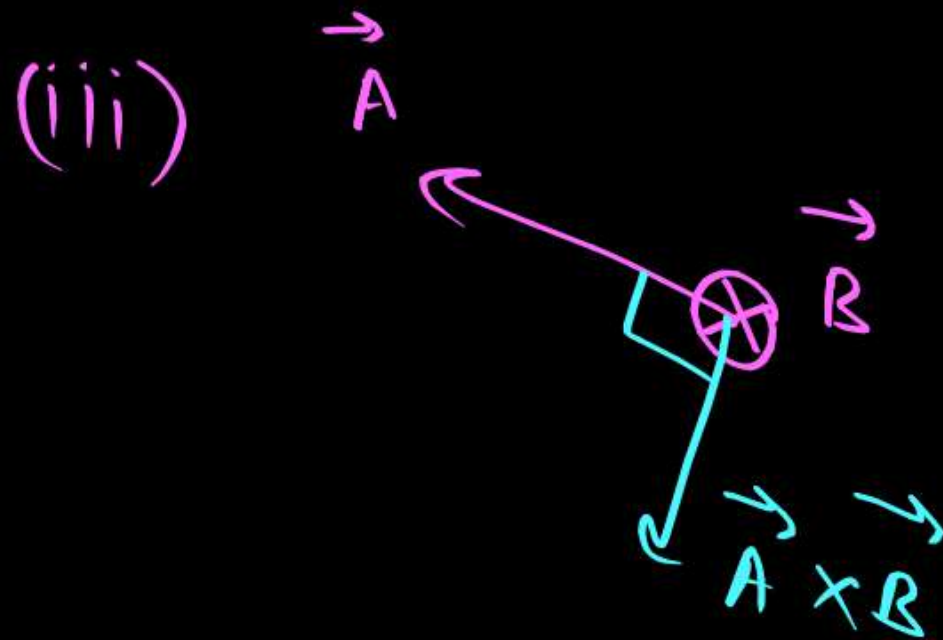
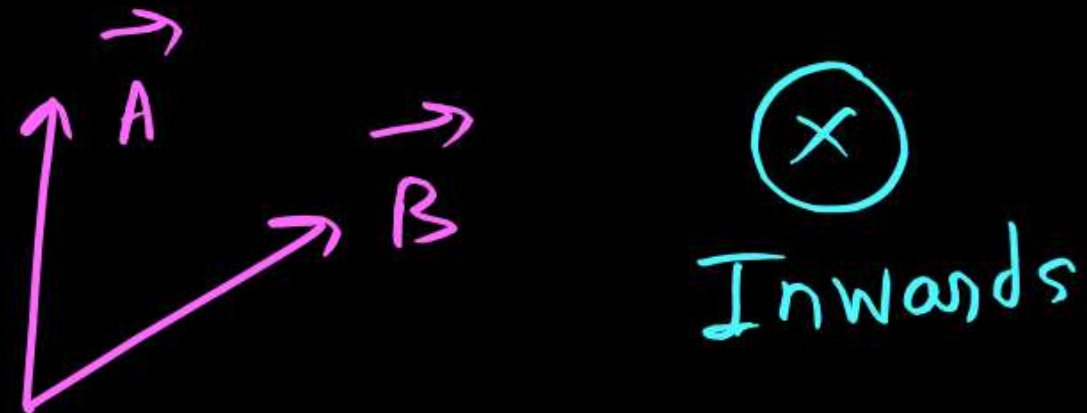


$$\vec{B} \times \vec{A} = -(\vec{A} \times \vec{B})$$

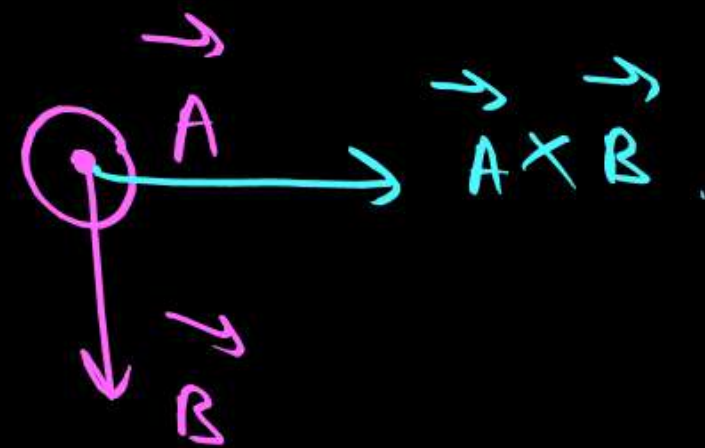
Q. Find the dirⁿ of $(\vec{A} \times \vec{B})$



(ii)



(iv)





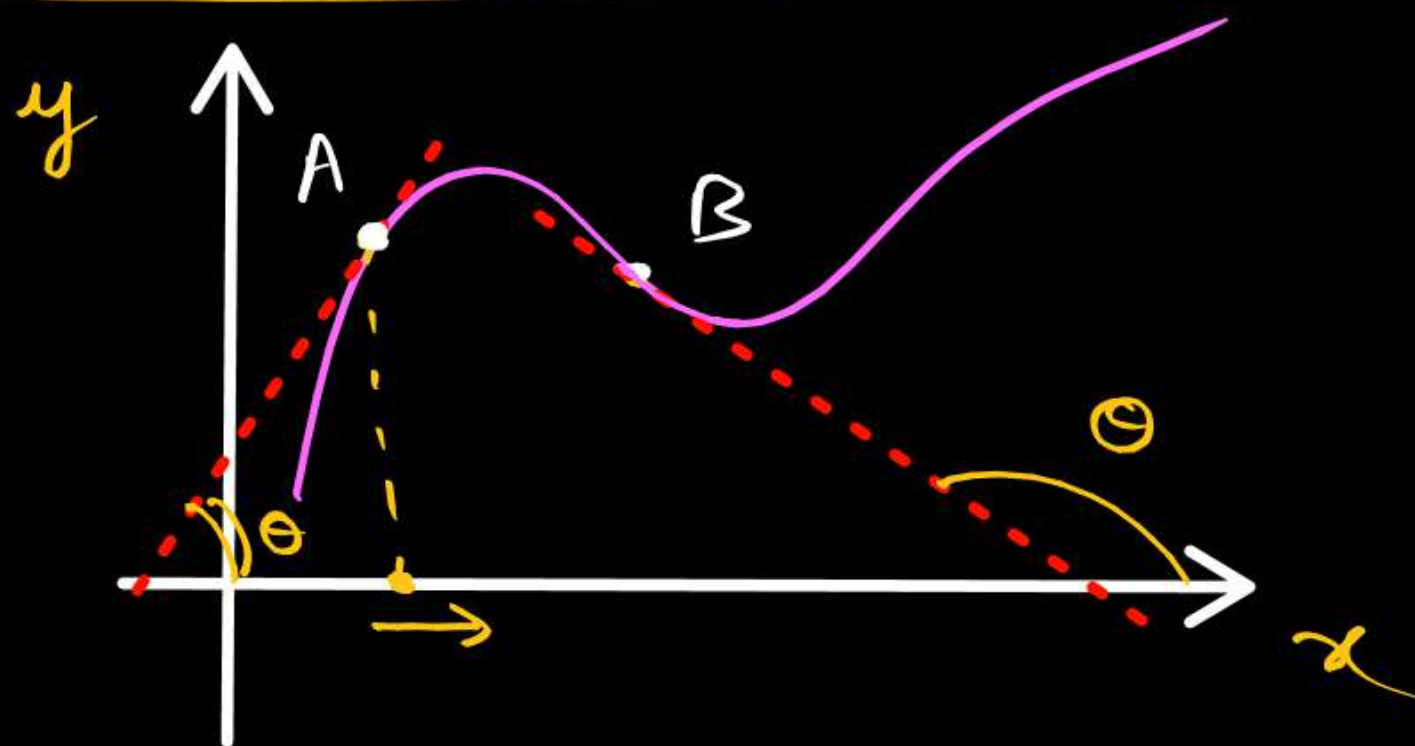
Differentiation

$\frac{dy}{dx}$ = rate of change
of y w.r.t x

(i) $\frac{d\vec{v}}{dt}$ = rate of change
of $\vec{v}$ w.r.t $t = \vec{a}$

(ii) $\vec{E} = -\frac{dV}{dr}(\hat{r})$ = rate of fall
of potential
w.r.t distance

Graphical Meaning:



$\frac{dy}{dx} = \tan \theta$ = Slope of tangent
at that point.

$\left(\frac{dy}{dx}\right)_A = +ve$ | $\left(\frac{dy}{dx}\right)_B = -ve$



Formulae



$$(1) \quad \frac{d}{dx} x^n = nx^{n-1}$$

$$(2) \quad \frac{d}{dx} \left(\frac{1}{x^n} \right) = -\frac{n}{x^{n+1}}$$

$$(3) \quad \frac{d}{dx} \sin x = \cos x$$

$$(4) \quad \frac{d}{dx} \cos x = -\sin x$$

$$(5) \quad \frac{d}{dx} \tan x = \sec^2 x$$

$$(6) \quad \frac{d}{dx} \sec x = \sec x \tan x$$

$$(7) \quad \frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$(8) \quad \frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$



Formulae

$$(9) \quad \frac{d}{dx} \log x = \frac{1}{x}, \text{ for } x > 0$$

$$(10) \quad \frac{d}{dx} e^x = e^x$$

$$(11) \quad \frac{d}{dx} a^x = a^x \log a, \text{ for } a > 0$$

$$(12) \quad \frac{d}{dx} \log_a x = \frac{1}{x \log a}, \text{ for } x > 0, a > 0, a \neq 1$$

(13)

$C = \text{const.}$

$$\frac{dC}{dx} = 0$$



Product Rule

$$y = f(x) \cdot g(x)$$

$$\frac{dy}{dx} = \frac{d[f(x)]}{dx} \cdot g(x) + f(x) \cdot \frac{d[g(x)]}{dx}$$

$$y = \underline{x} \underline{e^x}$$

$$\begin{aligned} \frac{dy}{dx} &= (1)e^x + x \cdot e^x \\ &= e^x(1+x) \end{aligned}$$

Question

$$(i) \quad \underset{\substack{\uparrow \\ 1}}{x} \cdot \underset{\substack{\uparrow \\ 2}}{\ln x} = y$$

$$\frac{dy}{dx} = 1 \cdot \ln x + x \cdot \frac{1}{x}$$

$$= \ln x + 1$$

$$(ii) \quad \overset{\uparrow 1}{x^2} \cdot \overset{\uparrow 2}{\sin x} = y$$

$$\frac{dy}{dx} = 2x \cdot \sin x + x^2 \cdot \cos x$$



Quotient Rule

$$y = \frac{f(x)}{g(x)} \Rightarrow \frac{dy}{dx} = \frac{\frac{d[f(x)]}{dx} \cdot g(x) - f(x) \cdot \frac{d[g(x)]}{dx}}{[g(x)]^2}$$

ex: $y = \frac{\overset{\rightarrow N}{\sin x}}{\underset{\rightarrow D}{x}} \Rightarrow \frac{dy}{dx} = \frac{\cos x \cdot x - \sin x \cdot (1)}{x^2}$

$$= \frac{x \cos x - \sin x}{x^2}$$

Question

$$(i) \frac{x^2}{x^3+1} = y$$

$$\frac{dy}{dx} = \frac{2x \cdot (x^3+1) - x^2 \cdot (3x^2+0)}{(x^3+1)^2}$$

$$= \frac{2x^4 + 2x - 3x^4}{(x^3+1)^2}$$

$$= \frac{-x^4 + 2x}{(x^3+1)^2}$$

$$(ii) \tan x = y = \frac{\overset{N}{\sin x}}{\underset{D}{\cos x}}$$

$$\frac{dy}{dx} = \frac{\cos x \cdot \cos x - \sin x [-\sin x]}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$= \sec^2 x$$



Chain Rule

$$(i) \quad y = \sin(x^2)$$

$$\begin{aligned} \frac{dy}{dx} &= \cos(x^2) \cdot 2x \\ &= 2x \cos(x^2) \end{aligned}$$

$$(ii) \quad y = \ln(\cos x)$$

$$\log x \rightarrow \frac{1}{x}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\cos x} \cdot (-\sin x) \cdot (1) \\ &= -\tan x \end{aligned}$$

$$y = \log[\cos(x^2)]$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\cos x^2} \cdot (-\sin x^2) \cdot 2x \\ &= -2x \cdot \tan x^2 \end{aligned}$$

Question

$a \rightarrow \text{const.}$

$$(iii) \quad y = \rightarrow \underline{\underline{(x^2 + a^2)^{3/2}}}$$

$$\frac{dy}{dx} = \frac{3}{2} (x^2 + a^2)^{1/2} \cdot [2x + 0]$$

$$= \cancel{2x} \cdot \frac{3}{\cancel{2}} (x^2 + a^2)^{1/2}$$

$$= 3x \cdot (x^2 + a^2)^{1/2}$$

$$(iv) \quad y = \frac{1}{\ln x} = \rightarrow (\ln(x))^{-1}$$

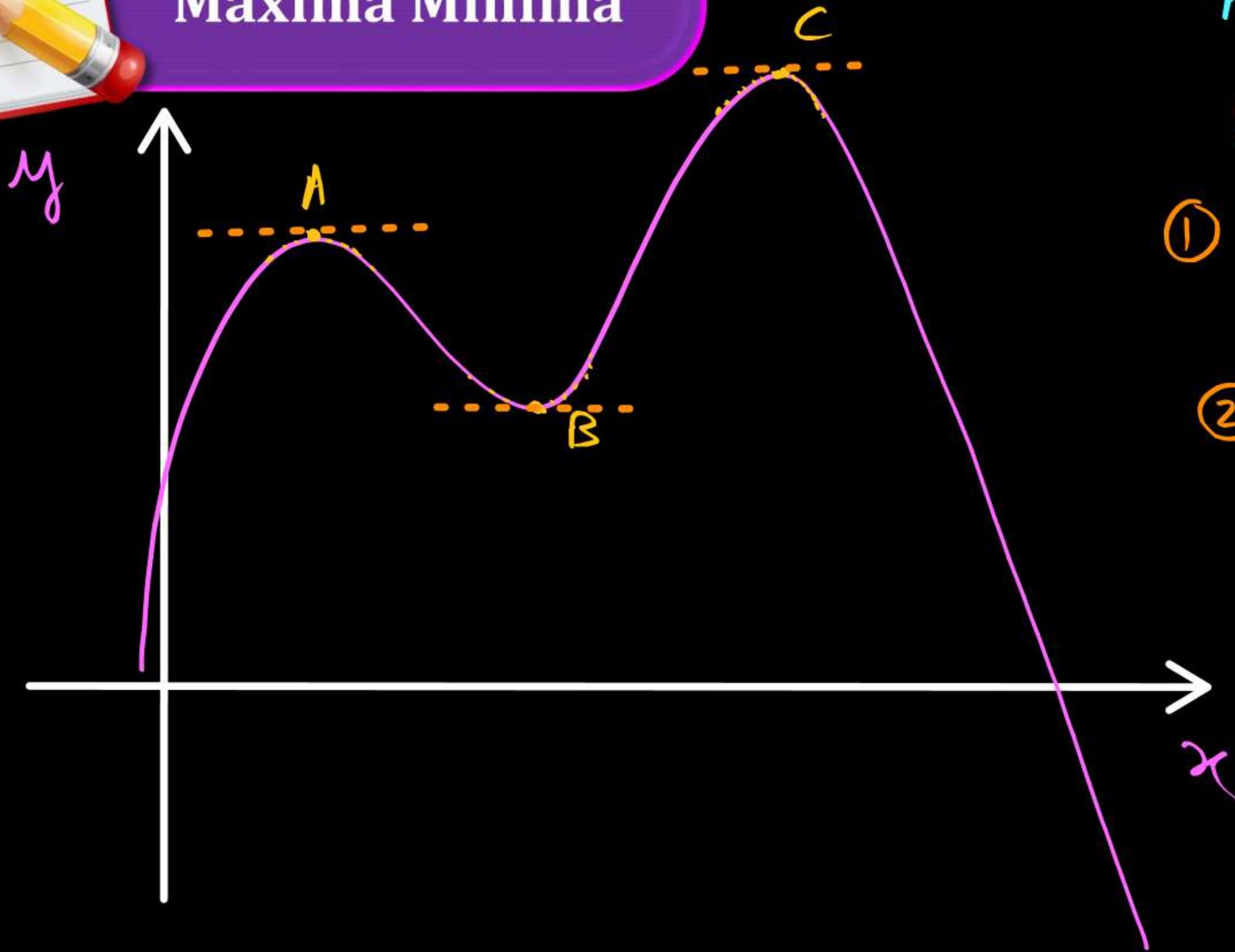
$$\frac{dy}{dx} = (-1)(\ln x)^{-2} \cdot \frac{1}{x}$$

$$\boxed{\frac{dy}{dx} = \frac{-1}{x (\ln x)^2}}$$

$$y = x^{-1}$$

$$\frac{dy}{dx} = (-1)x^{-2}$$

Maxima Minima



A, C $\rightarrow$ local maxima

B $\rightarrow$ local minima.

① $\frac{dy}{dx} = 0$

② Obtain the values of x .
(ex: $x = x_1, x_2, x_3, \dots$).

③ $\frac{d^2y}{dx^2}$ & substitute values of x obtained in step-2.

④ (i) If $\frac{d^2y}{dx^2} > 0 \Rightarrow y$ has a local minima
at that value of x .

(ii) If $\frac{d^2y}{dx^2} < 0 \Rightarrow y$ has a local maxima
at that value of x .

(iii) If $\frac{d^2y}{dx^2} = 0 \Rightarrow y$ neither has a local maxima
nor local minima at that value of x .

Question

$$y = \underline{x^3 - 3x^2 - 45x - 7}$$

$$\textcircled{1} \quad \frac{dy}{dx} = 3x^2 - 6x - 45 = 0$$

$$\Rightarrow 3x^2 - 9x + 15x - 45 = 0$$

$$\Rightarrow 3x(x-3) + 15(x-3) = 0$$

$$\Rightarrow (3x+15)(x-3) = 0$$

$$\Rightarrow \boxed{x = -5, 3}$$

$$\textcircled{2} \quad \frac{d^2y}{dx^2} = 6x - 6$$

$$\text{(i)} \quad x = -5 \Rightarrow \frac{d^2y}{dx^2} = 6(-5) - 6 = -36 < 0$$

$y \rightarrow$ local max., at $x = -5$

$$y_{\max} = (-5)^3 - 3(-5)^2 - 45(-5) - 7$$

$$\text{(ii)} \quad x = 3 \Rightarrow \frac{d^2y}{dx^2} = 6(3) - 6 = 12 > 0$$

$\Rightarrow y \rightarrow$ local min., at $x = 3$

Q. $y = x^2 - 6x + 5$
 $= (x-1)(x-5)$

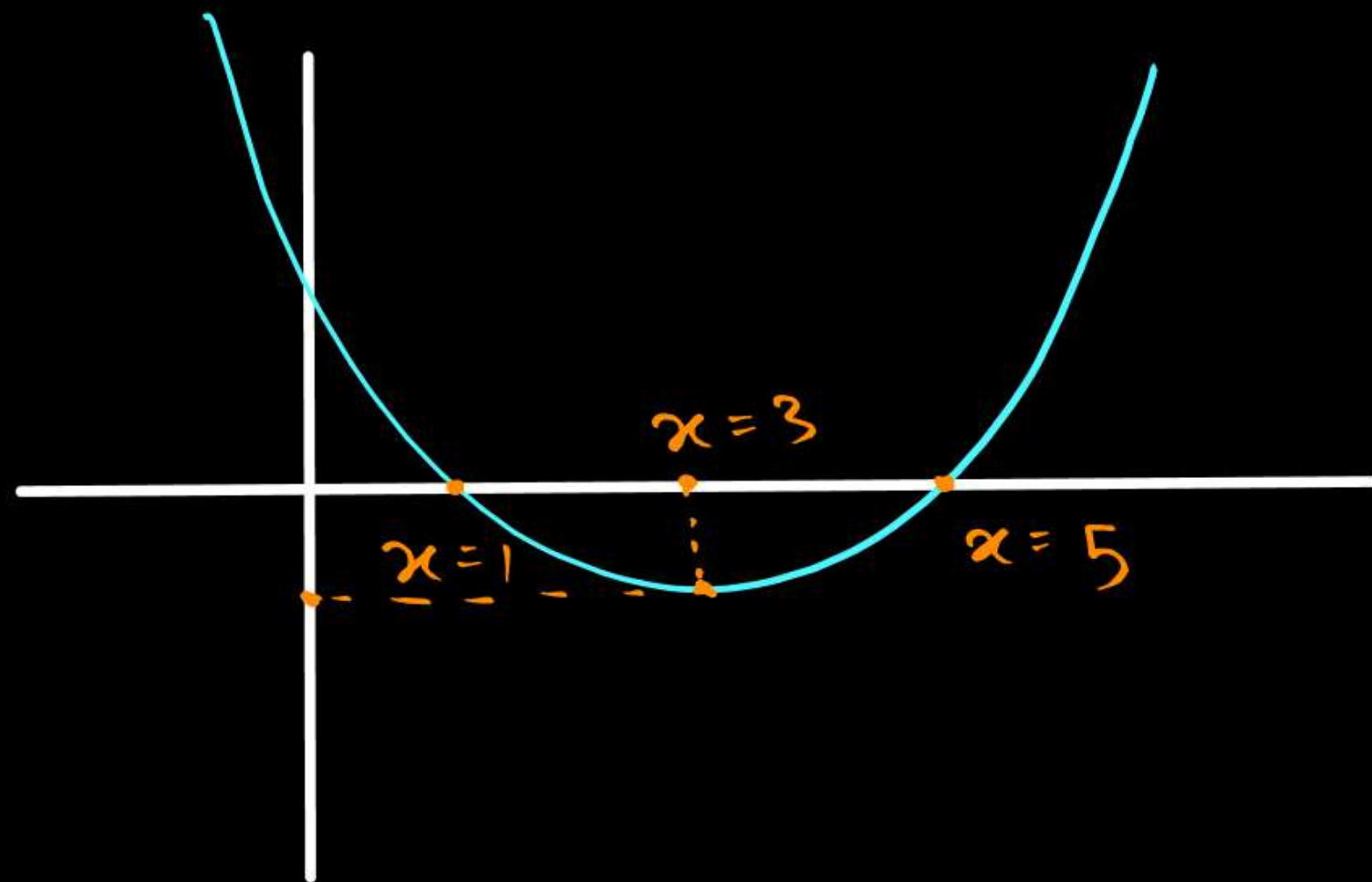
① $\frac{dy}{dx} = 2x - 6 + 0 = 0$

$\Rightarrow 2x = 6 \Rightarrow \boxed{x=3}$

② $\frac{d^2y}{dx^2} = 2 > 0$

$\Rightarrow$ local minima.

$y_{\min} = 3^2 - 6(3) + 5 = -4$



Q. $y = x^3$

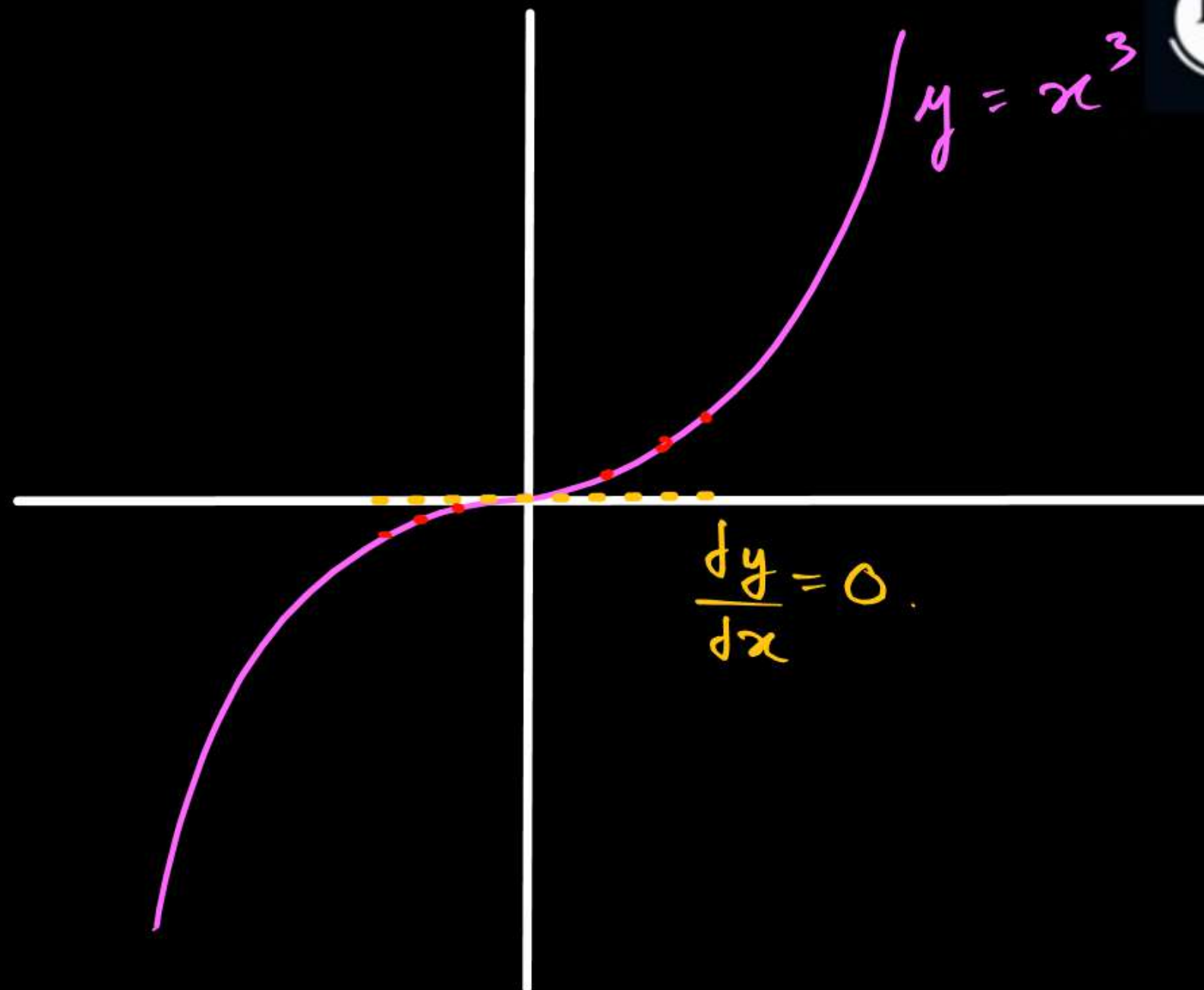
① $\frac{dy}{dx} = 3x^2 = 0$

$\Rightarrow \boxed{x=0} \star$

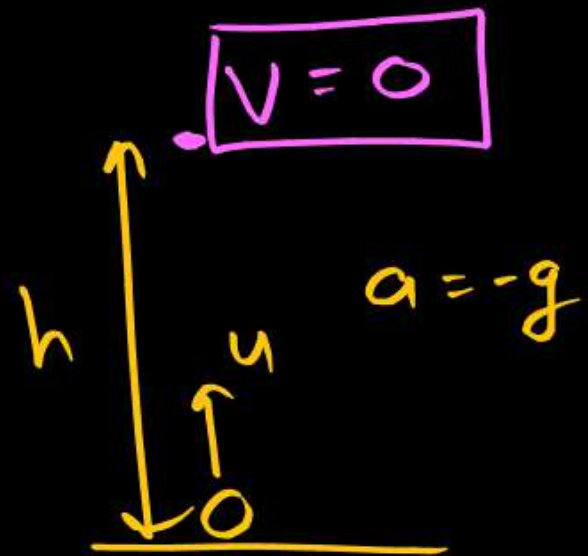
② $\frac{d^2y}{dx^2} = 6x$

③ $\boxed{x=0 \Rightarrow \frac{d^2y}{dx^2} = 0}$

$\rightarrow$ neither
max
nor
min



$$s = ut - \frac{1}{2}gt^2$$



$$\textcircled{1} \quad \frac{ds}{dt} = u - \cancel{\frac{1}{2}}g \cdot \cancel{2}t$$

$$= u - gt = 0$$

$$\Rightarrow \boxed{t = \frac{u}{g}}$$

$$\frac{ds}{dt} = 0$$

$$\Rightarrow \boxed{V=0}$$



Integration

Physically,

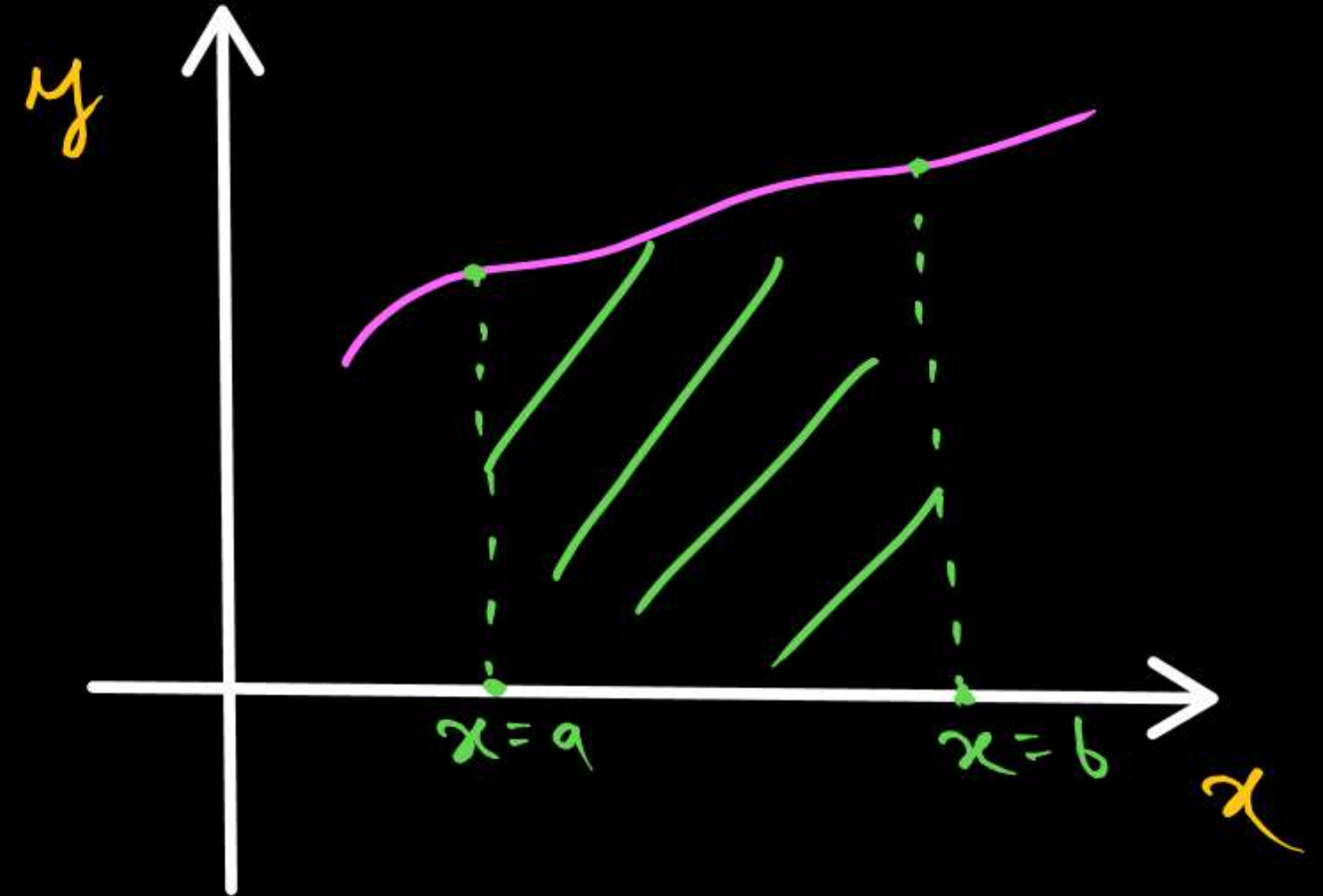
$$y = f(x).$$

$$\sum_{x=a}^b f(x) \Delta x$$

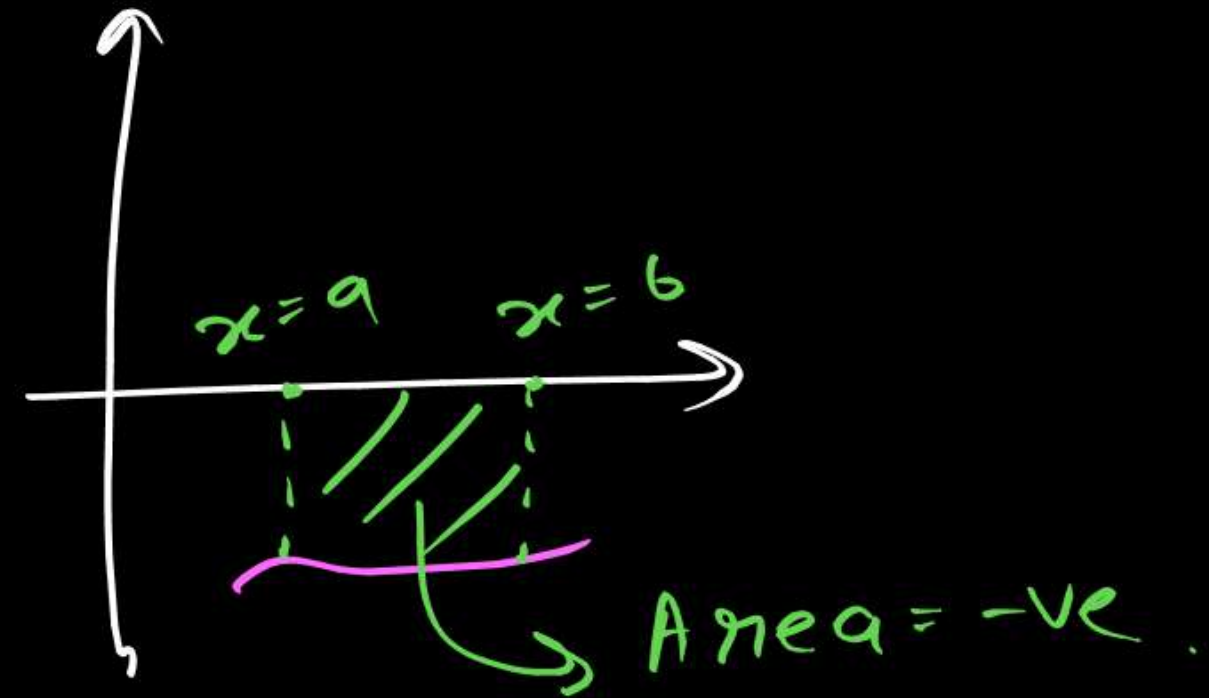
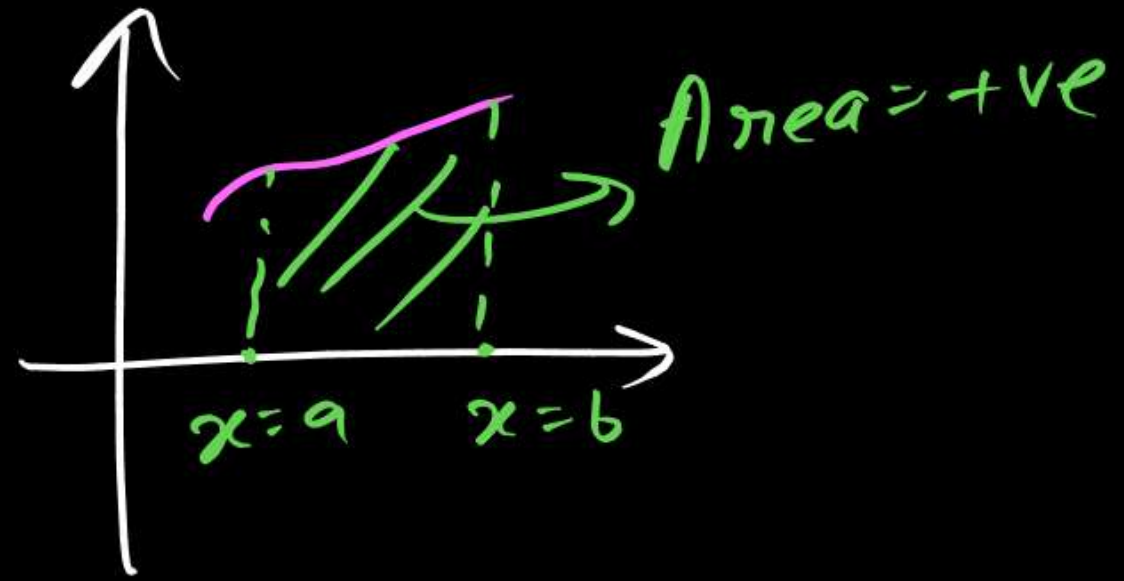
$$\rightarrow \int_a^b f(x) dx$$

= Sum of large no. of very small quantities.

Graphically,



$$\int_a^b f(x) dx = \text{Area under the curve.}$$



$$\int_a^b y \, dx$$

length



Formulae

$$(1) \quad \int x^n dx = \frac{x^{n+1}}{n+1} + c, n \neq -1$$

$$(2) \quad \int \frac{1}{x} dx = \log|x| + c$$

$$(3) \quad \int e^x dx = e^x + c$$

$$(4) \quad \int \sin x dx = -\cos x + c$$

$$(5) \quad \int \cos x dx = \sin x + c$$



Formulae

$$(6) \quad \int \sec^2 x \, dx = \tan x + c$$

$$(7) \quad \int \operatorname{cosec}^2 x \, dx = -\cot x + c$$

$$(8) \quad \int \sec x \tan x \, dx = \sec x + c$$

$$(9) \quad \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$$

$$(10) \quad \int 1 \, dx = x + c$$

$$\textcircled{1} \int \left(\frac{1}{\sqrt{x}} + \frac{1}{x^2} - 3x^{4/3} \right) dx = \int \left(x^{-1/2} + x^{-2} - 3x^{4/3} \right) dx.$$

$$\int x^n dx$$

$$= \frac{x^{n+1}}{n+1} + C.$$

$$= \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + \frac{x^{-2+1}}{-2+1} - \frac{3x^{\frac{4}{3}+1}}{\frac{4}{3}+1}$$

$$= \frac{x^{1/2}}{1/2} + \frac{x^{-1}}{-1} - \frac{3x^{7/3}}{7/3}$$

$$= 2\sqrt{x} - \frac{1}{x} - \frac{9x^{7/3}}{7} + C.$$

$$\textcircled{2} \int \frac{(x^3 - 1)}{x - 1} dx = \int \frac{\cancel{(x-1)} (x^2 + x + 1)}{\cancel{x-1}} dx$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + x$$

$$a^3 - b^3$$

$$= (a - b)(a^2 + ab + b^2)$$



Ax + B form

$$\int \frac{dx}{x} = \ln x$$



$$\textcircled{1} \int \cos(Ax+B) dx = \frac{1}{A} \sin(Ax+B)$$

$$\textcircled{2} \int \sin(Ax+B) dx = -\frac{1}{A} \cos(Ax+B)$$

$$\textcircled{3} \int (Ax+B)^n dx = \frac{1}{A} \cdot \frac{(Ax+B)^{n+1}}{n+1}$$

$$\textcircled{4} \int e^{(Ax+B)} dx$$

$$= \frac{1}{A} e^{(Ax+B)}$$

$$\textcircled{5} \int \frac{dx}{Ax+B} = \frac{1}{A} \ln(Ax+B)$$

Question

(i) $\int \sin(\omega t + \phi) dt$

Diagram showing the integration of $\sin(\omega t + \phi)$ with respect to t . The expression is enclosed in a dashed box. Above the box, the letters A, x , and B are written. Arrows point from A to the first part of the expression, from x to the argument of the sine function, and from B to the argument of the sine function.

$$= -\frac{1}{\omega} \cos(\omega t + \phi)$$

(ii) $\int \frac{dx}{\sqrt{4-2x}}$

Diagram showing the integration of $\frac{dx}{\sqrt{4-2x}}$ with respect to x . The expression is enclosed in a dashed box. Above the box, the letters A, x , and B are written. Arrows point from A to the first part of the expression, from x to the argument of the square root, and from B to the argument of the square root. The expression is also written as $\int (-2x+4)^{-1/2} dx$. The term $-1/2$ is circled and labeled with an arrow pointing to the exponent.

$$= \frac{1}{(-2)} \frac{(-2x+4)^{1/2}}{1/2}$$

$$= - (4-2x)^{1/2}$$

$$= -\sqrt{4-2x}$$



Definite Integration

$$\int f(x) dx = g(x)$$

b → upper limit

$$\int_a^b f(x) dx = [g(x)]_a^b$$

a → lower limit

$$= g(b) - g(a)$$

ex: (i) $\int_0^{\pi/2} \cos x dx = [\sin x + c]_0^{\pi/2}$

$$= \left(\sin \frac{\pi}{2} + \cancel{c} \right) - \left(\sin 0 + \cancel{c} \right)$$

$$= 1 - 0 = 1$$

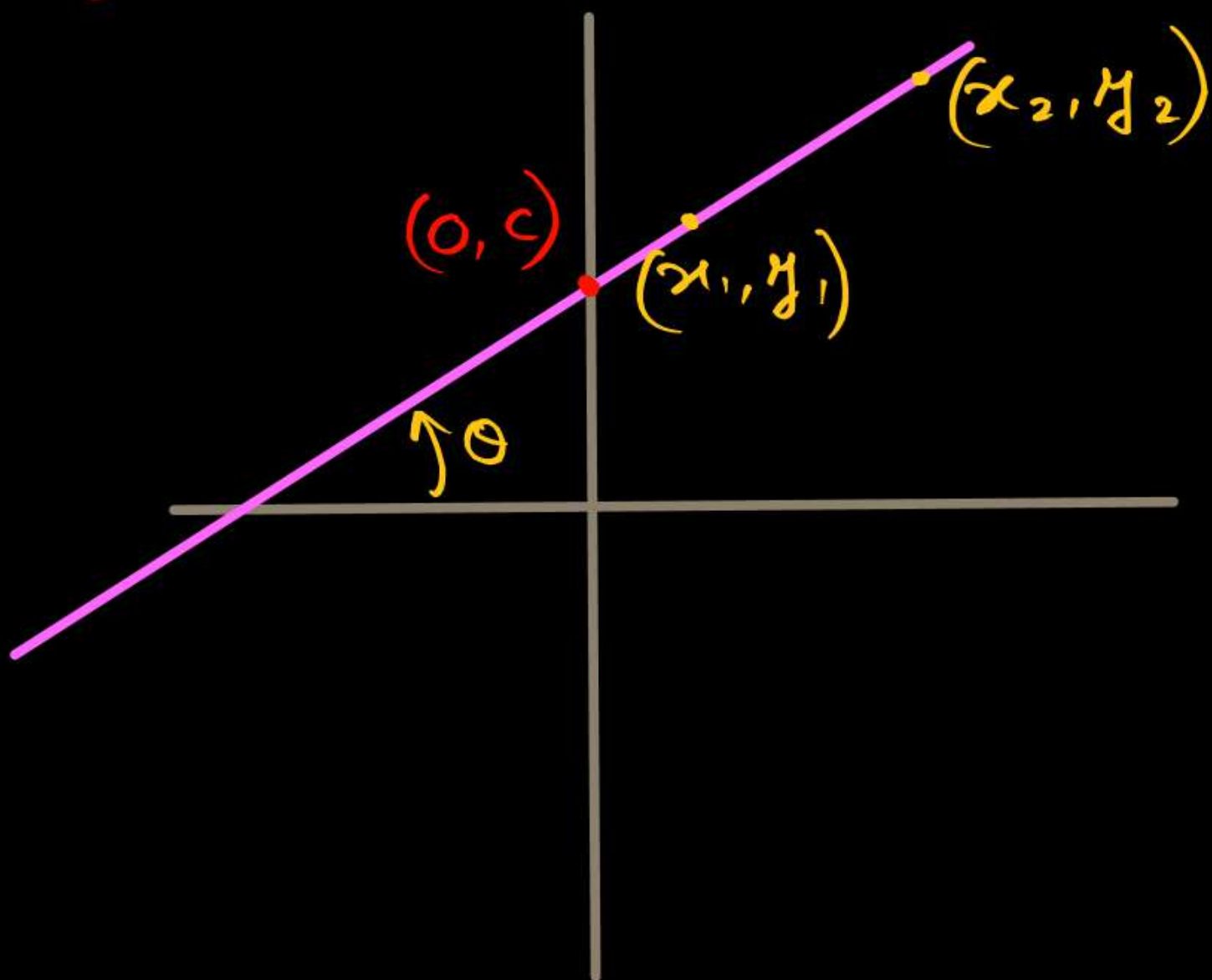
Question





Graphs

(Straight lines)



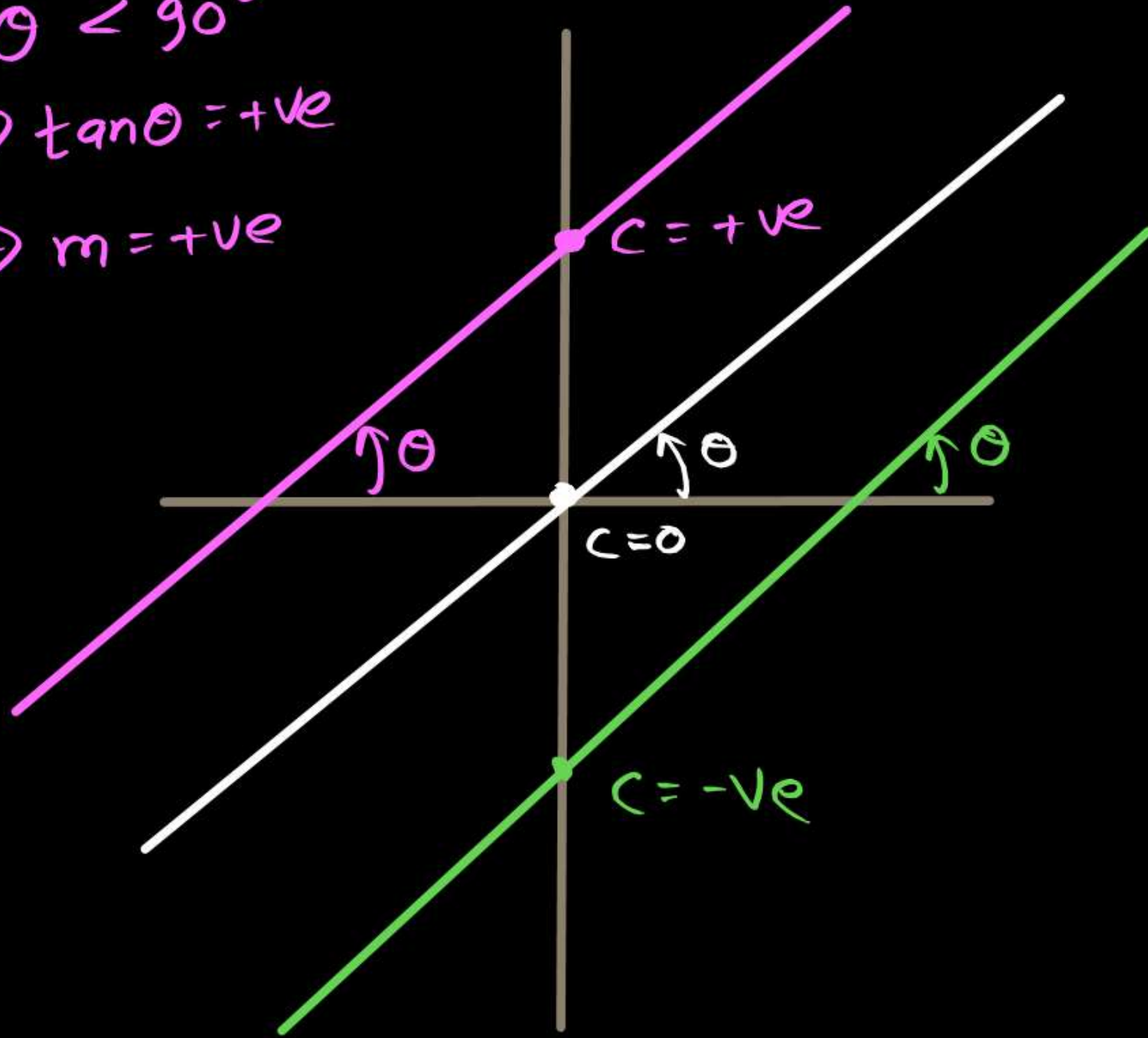
$$y = mx + c$$

$$m = \text{slope} = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$$

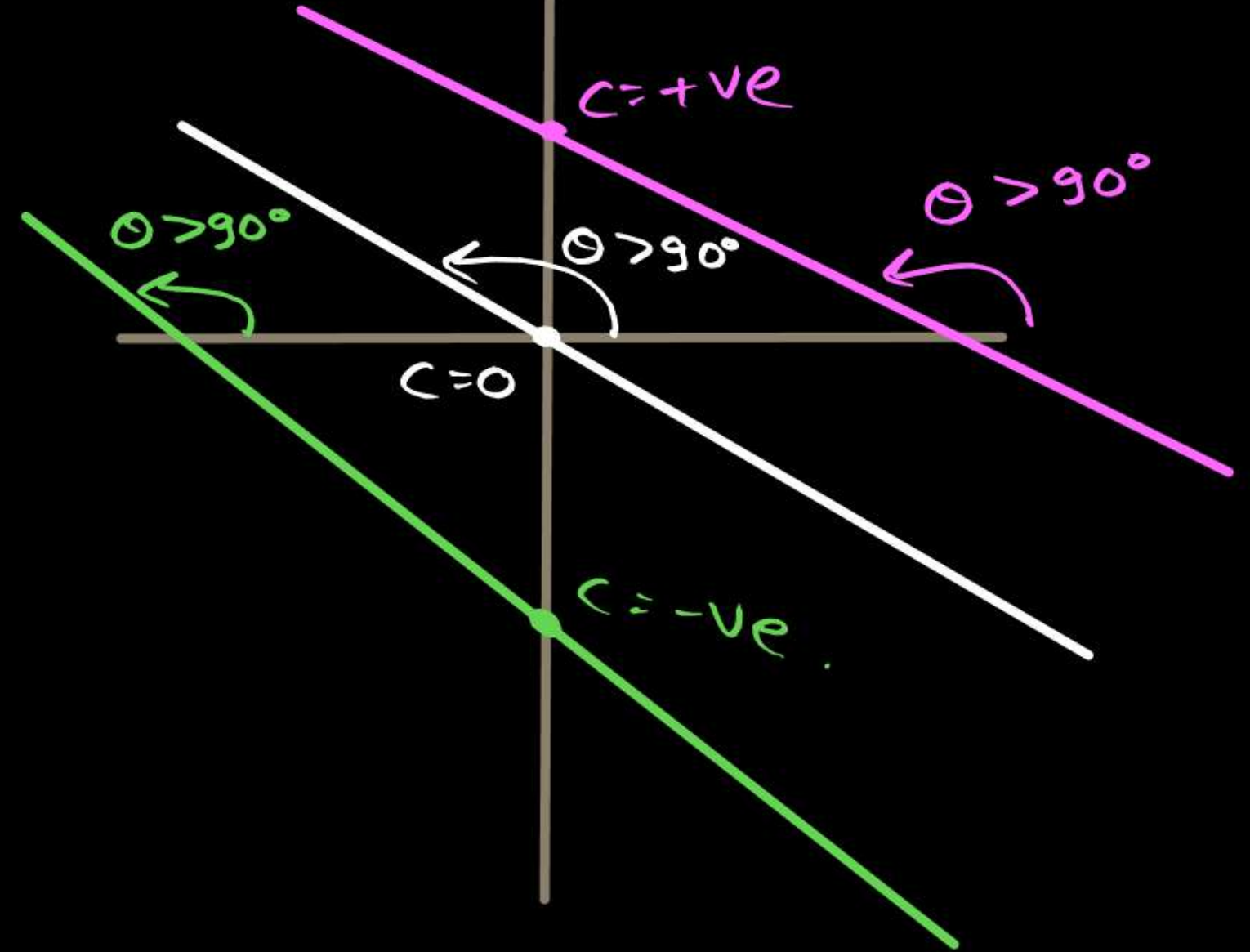
$c \rightarrow$ y-intercept

(y co-ordinate of pt. where
line intersects y-axis)

$\theta < 90^\circ$
 $\Rightarrow \tan \theta = +ve$
 $\Rightarrow m = +ve$



$\theta > 90^\circ$
 $m = \tan \theta < 0$





Homework



★ Diff., Integ., Max-min, Vector (S.C.Q.)
on (Integer type)

THANK YOU

